

SUMMARY

The research reported in this article was carried out as part of a larger project investigating the impact of the US Space Program on the American economy. The author describes the influence of governmental programs in space and defense on the growth of the computer and semiconductor industries. Three types of economic impacts are analyzed: the role of the federal government in financing research and development; the importance of the government as a market, particularly in the early stages of an industry's growth; and the extent to which government policy encouraged the entrance of new firms and permitted the survival of companies that might otherwise have failed. The paper also deals with the contribution of Defense and NASA to the acceleration of technological changes and the development of specialized manpower on which the progress of the industry largely depended.

The author concludes that large government programs can play an essential role in the growth of high technology industries. The significance of space-defense programs for US semiconductor and computer firms can only be perceived by reviewing each type of impact — economic, technological and manpower. The government's early R & D support, assured demand during start-up periods, encouragement of competition, acceleration of technological progress, and its facilitating of professional mobility, are the separate impacts which together greatly aided US firms in their rise to a dominant position in the world semiconductor and computer industries.

government programs and the growth of high-technology industries

by

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INTRODUCTION

As the activities and expenditures of the federal government have expanded during the past two decades, the impacts of large public programs have come under closer scrutiny. Impacts are distinct from the specific accomplishments of an agency in carrying out its overall mission. The placing of a man on the moon, or the orbiting of an earth resources satellite are examples of a specific accomplishment. Whereas specific accomplishments relate directly to agency objectives, impacts do not. In achieving a specific goal, a public agency may stimulate the development of a new technology or a new industry, dramatically alter the infrastructure of a city or a region of the country, or create new occupational categories.

These ancillary effects of large public programs are exemplified by the role of large US government defense and space programs in nurturing the development of two high-technology industries – computers and semiconductors. American firms have dominated the semiconductor (solid-state electronic components) and computer industries since their post-World War II origins. IBM, for example, accounts for over two-thirds of all computer installations both inside and outside of the United States. In semiconductors, American firms have carried out all of the important innovations since Bell Laboratories discovered the transistor in 1948 [1]. This paper identifies and describes three types of influence – economic, technological, and manpower – which the space and defense programs exerted on these new industries [2]. An analysis of all three impact categories will provide insight into the full range of support which a public program can provide to a high-technology industry.

ECONOMIC IMPACTS: SUPPORT FOR RESEARCH AND DEVELOPMENT

The economic dimension of government influence has been evident in both the computer and semiconductor industries. Specifically, three types of economic impacts can be identified:

- (1) Direct and indirect financial support by space—defense programs for semiconductor and computer R & D,
- (2) Assured demand provided by space and defense programs during the early years of the industry,
- (3) The use of space and defense demand to support new firms and to affect the competitive balance within the industry as it matures.

Because innovation has been so critical in the development of the computer and semiconductor industries we shall begin by considering space—defense financial support for research and development.

Prior to the discovery of the transistor by Bell Laboratories in 1948, European firms were the equal of American firms in advancing component technology. Some of the most important advances in electron tubes had, in fact, been made by European firms and European universities. Beginning with the transistor, almost all the important inventions and innovations in components have been made in the United States, and there has been a lag of one to four years before manufacture has begun in Europe, often by American subsidiaries [3].

The leading role which American firms have played in advancing component technology is directly related to the R & D support provided by the government. In the early 1950's, the military services became convinced of the military importance of transistors and other semiconductor devices. They believed these devices could miniaturize, and thus make more portable, much military equipment and also improve the reliability of equipment in field operations. As a result, the Department of Defense financed semiconductor R & D on a large scale and aided the principal firms in enlarging their production capacity. By 1953, the Army Signal Corps was financing pilot production lines for transistors and related devices at five sites operated by Western Electric, General Electric, Raytheon, RCA, and Sylvania. Between 1955 and 1961, direct government funding for semiconductor R & D and production refinement totalled \$66 million [4].

This government financial support was provided in two forms: (1) direct contracts for R & D projects and (2) government-financed production refinement programs. Approximately 30% of the funds were expended in 1956 when a special appropriation of \$14 million was allocated to production refinement projects for transistors. Contracts for the development of thirty

different types of transistors over the following several years were placed with 12 semiconductor companies. In many cases the DOD investment was matched by similar amounts of capital equipment or plant space supplied by the contracting companies. Contracts called for the delivery of about three thousand transistors of each type to be made on production lines capable of producing that many per month. Eventually the special \$14 million expenditure aided in greatly expanding semiconductor production capacity in the US [5]. The Army Signal Corps was extremely active in supporting semiconductor research during this period because of its concern with improving military communications equipment. In 1961, this agency employed 27 commercial contractors on research programs for semiconductor devices, 40 commercial firms on electronic parts and materials projects, and 13 commercial firms on quartz crystal units. The Signal Corps program produced some important commercial benefits for these firms by enabling them to substantially reduce the length of commercial development projects [6].

The \$66 million of direct government funding on semiconductor R & D significantly understate the amount of total government support since many semiconductor firms received support indirectly. Government contracts for space and weapons systems enabled the prime contractors to subcontract with semiconductor firms for R & D. A 1960 DOD survey measured the amount of direct federal semiconductor support as well as the indirect amount which was subcontracted to semiconductor firms by weapons systems prime contractors. The survey estimated that DOD direct and indirect semiconductor R & D funding totaled \$13.9 million and \$16.2 million in 1958 and 1959, respectively. These totals were more than double the amount of direct DOD R & D funding in the respective years. The survey also revealed the relative importance of government-sponsored R & D; in 1958 and 1959, the Department of Defense funded 25% and 23%, respectively, of total semiconductor industry R & D expenditures [7]. As these figures make clear, substantial support was provided for semiconductor R & D during the early stages of the industry's development. The \$66 million in direct support, as well as nearly equal amounts in indirect support, aided the firms in refining their products and expanding production capacity.

A similar pattern of early large-scale government R & D support is evident in the computer industry. Federal agencies began to play an instrumental role in the development of the US computer industry in the 1940's. Although the first US computer, the Automatic Sequence Controlled Calculator (ASCC), was developed by a Harvard-IBM team between 1937 and 1944, all subsequent US computer developments from the end of World War II, until the mid-1950's, had the strong financial support of various US government agen-

cies. Work on the first electronic computer, the Electronic Numerical Integrator and Calculator (ENIAC), began at the University of Pennsylvania in 1942 and was completed in 1946. ENIAC received financial support from the Army and was mainly designed for calculating trajectories of shells and bombs. From 1945 to 1955, the US Army, Navy and Air Force, the Atomic Energy Commission, and the National Bureau of Standards all placed major contracts for the development of improved computers with universities and with the first firms who began design and manufacture – especially Remington Rand (Univac) and, later, IBM. As a result, rapid progress was made in solving some of the problems of logic design, memory storage systems and programming techniques.

The major technical advances during this period were made in the large computers demanded by the military and other government agencies. These advances were subsequently incorporated in the medium and small computers, which were designed for smaller scale, less complex applications. In the 1950's, government contract R & D accounted for about 60% of total R & D expenditures at IBM; by the early 1960's, government-sponsored R & D was a very small part of the company's \$125 million R & D budget. The very large IBM computers, such as STRETCH, were built on contract for government agencies, but the highly successful transistorized 1401 series and the 360 and 370 series have been privately funded projects.

Government support of US computer developments takes on added significance when the US experience is contrasted with that of other countries. Although it is widely believed that the first successful computers were built in the United States and Great Britain, the first electro-mechanical computer was actually developed by Frederick Zuse in Berlin between 1936 and 1941. Moreover, Zuse was also the first to begin work on the electronic computer during the early 1940's. But the lack of official support by the German government for his project and the disruption which occurred at the end of the War caused the lead in computer development to pass to the United States, and to a lesser extent, to Britain.

In Britain, electronic computer development work began in 1946–47 after British scientists and engineers had visited the first American installations. The earliest work was at London, Manchester, and Cambridge Universities, and at the Elliott Brothers' Research Laboratory on digital computing for fire control for the Admiralty. Although financial support for several computer projects was provided by the National Research and Development Corporation (NRDC), total funding was limited to \$2.5 million during the 1947–57 period because British military agencies remained unconvinced that there were substantial benefits to be derived. Consequently, British computer firms

were never able to compete effectively with their US counterparts. Initially, ten British firms were involved in the development and manufacture of computers. But few of these firms ever developed a computer model with sales exceeding fifty units, and by the mid-1960's the number of British computer manufacturers had declined from ten to three [8].

THE IMPACT OF ASSURED DEMAND

The preceding discussion makes it clear that government R & D funding provided vital financial support for the early technical development of semiconductors and computers. Government programs were equally important when these products first reached the market; the earliest sales of semiconductors and computers were to military agencies. The first semiconductors were marketed in the early 1950's. By 1955, the semiconductor market was \$40 million and military agencies accounted for 38% of these sales. The space-defense market for semiconductors grew from \$15 million in 1955 to \$222 million in 1961. At times, space-defense demand accounted for almost 50% of the total semiconductor market (table 1) [9].

Government demand proved even more significant during the first years of integrated circuit commercial production. The earliest integrated circuits were used in the Minuteman missile program and various Navy programs in 1962 and 1963. In 1962, space-defense sales accounted for the total \$4 million integrated circuit market. By 1968, the space-defense share of integrated circuits had declined to 37% (table 1).

Space-defense business was vital in an industry where learning economies are substantial. Typically, the price for each new component innovation was too expensive for industrial or consumer electronic products. But NASA and DOD provided a substantial market for new devices, because reliability and performance received higher priority over costs in the space-defense environment. Later, as the firms increased production, learning economies were realized and costs fell. In a few years, the price was low enough to penetrate the industrial market, and eventually the consumer market.

Semiconductor firms estimate that the average cost of producing a new component falls between 20 and 30% every time its cumulative past production doubles. The impact of learning economies on the price of integrated circuits has been dramatic. In 1962, when the total market was \$4 million, the average price of an integrated circuit was approximately \$50. By 1968, the market had grown to \$312 million and the average price had dropped to \$2. At the 1973 price of \$.63, integrated circuit customers could purchase 50 units of

Table 1
US production of semiconductors and integrated circuits for defense requirements, 1955–68a)

Year	Semiconductors		Integrated circuits			Semiconductors and integrated circuits		
	Total ^{b)} production	Defense ^{b)} production	Defense as % of total	Total ^{b)} production	Defense ^{b)} production	Defense as % of total	Total ^{b)} production	Defense ^{b)} production
1955	40	15	38	d) —	—	—	40	15
1956	90	32	36	—	—	—	90	32
1957	151	54	36	—	—	—	151	54
1958	210	81	39	—	—	—	210	81
1959	396	180	45	—	—	—	396	180
1960	542	258	39	—	—	—	542	258
1961	565	222	39	—	—	—	565	222
1962	571	219	38	4 ^{c)}	4	100 ^{c)}	575	223
1963	594	196	33	16	15	94 ^{c)}	610	211
1964	635	157	25	41	35	85 ^{c)}	676	192
1965	805	190	24	79	57	72	884	247
1966	975	219	22	148	79	53	1,123	298
1967	879	205	23	228	98	43	1,107	303
1968	847	179	21	312	115	37	1,159	294

Sources: Data for semiconductors are from US Department of Commerce, Business and Defense Services Administration (BDSA), *Electronic Components: Production and Related Data*, 1952–1959 (1960); Data for 1960–1968 are from BDSA, Consolidated Tabulation: Shipments of Selected Electronic Components (annual reports) and *Electronic Industries Yearbook*, 1969 (Washington: Electronic Industries Association, 1969), table 55. These data appear in a different form in Tilton, *op. cit.*, Tables 4–7 and 4–8.

a) Defense production includes devices produced for Department of Defense (DOD), Atomic Energy Commission (AEC), Central Intelligence Agency (CIA), Federal Aviation Agency (FAA), and National Aeronautics and Space Administration (NASA) equipment.

b) All figures in millions of dollars.

c) Estimated.

d) Commercial production of integrated circuits began in 1962.

increased complexity for the same price they had paid for one unit in 1963 [10].

Computer manufacturers, as well as semiconductor firms, benefited from government purchases during the industry's early years. Military demand for computers was critical in the early 1950's. By the time Sperry Rand delivered the first UNIVAC computer designed for business data processing to General Electric in 1954, the military market for data processing systems had already reached nearly \$200 million. Although sales of commercial data processing hardware rose rapidly to \$500 million in 1960, the military and space market for data processing systems was still larger (\$900 million) in that year. The sales of commercial data processing systems did not surpass military and space hardware sales until 1962, nearly eight years after delivery of the first computer designed for business use (table 2).

Table 2
Computer market statistics, 1954–63 (millions of dollars)

Year	Commercial computer sales	Space–defense computer sales	Total computer sales	Sale of computer services	Sales of computers & services	Space–defense share of computers	Space–defense share of comp. & serv.
1954	–	200	200	–	200	100%	100%
1955	68	250	318	50	368	79%	54%
1956	185	300	485	175	660	62%	45%
1957	230	350	580	200	780	60%	45%
1958	200	500	700	250	950	71%	53%
1959	310	800	1110	300	1410	72%	57%
1960	600	900	1500	500	2000	60%	45%
1961	820	1000	1820	750	2570	55%	39%
1962	1200	1100	2300	1000	3300	48%	33%
1963	1370	1200	2570	1200	3770	47%	32%

Source: G. Burck et al., *The Computer Age* (Harper & Row, New York, 1965) pp. 73–78; A. Gropelli, *op. cit.*, appendix.

The relationship between military–space demand for computers and overall market growth is similar to that observed in the semiconductor industry. In both cases, space–defense demand accounted for a large share of the total market at the outset. The space–defense share of the total computer hardware market was initially 100% and remained over 60% for the next seven years. Although the space–defense share of the computer services market was lower than the space–defense share of the computer hardware market, government purchases still accounted for nearly half of total sales through 1960 and did not drop below one-third of the total market until 1963 [1].

The above statistics document the direct spending of space–defense agencies on computers, but they do not capture the indirect effect of space–defense expenditures resulting from computer spending by large aerospace companies. A computer science professor described the indirect effects of space–defense programs on the computer industry in this way [12]:

Big spenders are significant factors in the development of the American computer industry. American dominance of computers and computer science is largely due to massive government space—defense spending during the 1950–1965 period. These funds flowed to large aerospace companies who, in turn, became big spenders. The emergence of the pioneering IBM 7090 and 7094 lines was the result of these several “big spenders”.

The large space—defense market for computers produced three important secondary benefits for computer firms. First, space—defense sales enabled companies to fund higher levels of research and development which, in turn, promoted technological progress. Research advances, together with the learning economies which resulted from increasing volume levels, lead to continual declines in prices. An index of computer prices for the 1954–65 period revealed that prices in 1965 were approximately one-tenth those in 1954 (ref. [13], table 2.5).

A second important benefit provided by space—defense demand was experience with new types of computer applications. Military requirements for space, speed, and reliability resulted in more sophisticated applications than those first sought by business firms. Typically, the techniques utilized in a particular computer application would initially be developed and refined in a military system. Subsequently, the same techniques would be applied to a commercial data processing system. The most notable example of applying military computer experience to commercial problems is in the area of real-time computer systems. The first commercial real-time computer systems were descendants of the Air Force’s pioneering real-time air defense computer system — Semi-Automatic Ground Environment or SAGE. A computer company executive with substantial experience in both the space—defense and commercial markets stressed the relationship between applications in the two markets [14]:

Space and defense computer applications have served as a “crystal ball” for predicting the future direction of computer use in industry. Because many computer applications in the space and defense programs are on the leading edge of the technology, they point the way to future applications in industry (and elsewhere in government) two to three years hence.

A third benefit of space—defense demand was the “demonstration effect.” Sales in the commercial computer market depended on the acceptance of complex systems by business executives. Many of these executives lacked the technical knowledge needed to fully appreciate the potential of computers.

The combination of unfamiliarity and skepticism among business executives would have been a significant barrier to commercial sales were it not for the “demonstration effect” of space—defense computer systems. In the 1950’s military demonstrations of computer ability accelerated the business world’s acceptance of the computer. More recently, the widely publicized use of computers for NASA space missions during the 1960’s produced an unparalleled demonstration effect. One computer industry observer noted the significance of NASA’s use of computers [15] :

NASA’s successful use of computers, particularly on Apollo, greatly increased industry’s receptivity to computers. Managers became more willing to let computers control things, such as in process control of oil refineries. The momentum generated by Apollo had a big effect on confidence, producing a sudden turnaround in industry receptivity.

COMPETITION AND THE SUPPORT OF NEW FIRMS

Space—defense demand has not only stimulated the growth of the semiconductor and computer industries, it has also helped stimulate and maintain competition in these fields. The ability of new firms to enter the market is an essential characteristic of a competitive industry. In both the computer and semiconductor industries government demand has been a factor in supporting new firms, but space—defense stimulation of competition is more pronounced in the semiconductor market.

The first commercial producers of semiconductors were Western Electric and major tube manufacturers, such as General Electric, RCA, and Sylvania. General Electric, Raytheon, and RCA began producing transistors in 1951, and by 1954 eight different firms were manufacturing transistors. Other firms were slower to respond to the new technology, but they entered the market in large numbers during the mid-1950’s. The three types of firms which entered the market were: (a) firms which were large and diversified before they entered the semiconductor industry (Hughes, Motorola, and IBM); (b) small firms engaged in other branches of electronics or in other industries (Texas Instruments, General Instrument); and (c) new companies which were initially established to manufacture semiconductors (Transitron, National Semiconductor) [16].

Many of the new firms were able to enter the market because the military agencies were willing to buy from new and untried firms. By 1957 these new firms accounted for 64% of the total market and 69% of the government

Table 3
US semiconductor market shares of the major firms, selected years,
1957–66^{a)}

Type and name of firm	Percentage of market			
	1957	1960	1963	1966
Western Electric	5	5	7	9
<u>Receiving tube firms</u>				
General Electric	9	8	8	8
RCA	6	7	5	7
Raytheon	5	4	b)	b)
Sylvania	4	3	b)	b)
Philco-Ford	3	6	4	3
Westinghouse	2	6	4	5
Others	<u>2</u>	<u>1</u>	<u>4</u>	<u>3</u>
Subtotal	31	35	25	26
<u>New firms</u>				
Texas Instruments	20	20	18	17
Transitron	12	9	3	3
Hughes	11	5	b)	b)
Motorola	b)	5	10	12
Fairchild	b)	5	9	13
Thompson Ramo Wooldridge	b)	b)	4	b)
General Instrument	b)	b)	b)	4
Others	<u>21</u>	<u>16</u>	<u>24</u>	<u>12</u>
Subtotal	64	60	68	65
Total	100	100	100	100

Source: J. E. Tilton, *International Diffusion of Technology: The Case of Semiconductors* (The Brookings Institution, Washington D.C., 1971) p. 66.

a) Market shares are based on company shipments and include in-house and government sales.

b) Not one of the top semiconductor firms for this year. Its market share is included under "Others."

market (table 3). The two largest firms in the market, Texas Instruments and Transitron, with 20% and 11% market shares, respectively, had both benefited greatly from military demand. In early 1953, the military authorized the use of Transitron's gold-bonded diode before the company had made any significant commercial sales [17]. In 1954, Texas Instruments, building on research at Bell Laboratories, introduced the first silicon transistor. The innovation opened up a vast new market, particularly in products for military use.

Space program demand has been especially important in establishing a competitive market for metal-oxide-semiconductor (MOS) devices. The first MOS transistor was introduced by Fairchild in 1960. However, as a result of several technical difficulties and the departure of many MOS specialists, Fairchild dropped MOS devices from their product line after two years of commercial production. Then, beginning in 1963 the new firms quickly entered the field. General Micro Electronics (GME) was founded by Fairchild personnel in 1963 and received one of its first contracts to produce MOS integrated circuits from NASA. When Philco-Ford acquired GME in 1966 several GME employees left to form American Micro-Systems (AMI) which was subsequently awarded a NASA contract. The firm developed the technology required to produce stable MOS devices, and became one of the early suppliers of specialized commercial MOS devices. AMI eventually became a major source of such devices throughout the country [18].

The impact of space—defense demand on competition among computer firms is evident particularly in the industry's early years. Sperry Rand, the company which delivered the first UNIVAC computer, had a headstart on IBM during the early 1950's. IBM did not enter the electronic computer industry until 1953, but it was the clear industry leader by 1956. Most analyses attribute the overwhelming IBM success to the firm's aggressive salesmanship. While essentially correct, these analyses ignore the significance of space—defense demand for IBM. Space—defense purchases exceeded 30% of the firm's computer revenues in the mid-1950's and did not drop below 20% until 1958 [19].

During the late 1950's space—defense sales were vital to the growth of many new computer firms. The most significant of those new firms was the Control Data Corporation begun by former Sperry Rand employees in 1957 as a small but successful supplier of standardized systems for military and scientific use. Control Data raised revenues from practically nothing in 1958 to around \$121 million in the fiscal year ending in June 1964. By 1965 the firm had displaced RCA as the third largest computer manufacturer behind IBM and Sperry Rand. Space—defense business provided the initial impetus for Control Data's growth; almost all of the company's pre-1960's sales were to government agencies [20].

During the 1960–70 decade, space—defense sales aided several computer manufacturers, particularly Sperry Rand and Control Data, to maintain or improve their competitive position vis-à-vis IBM. A 1962 government policy shift from computer rental to the direct purchase of computers promoted the growth of IBM's competitors. IBM began to feel the impact of this policy when several federal agencies converted from IBM to other equipment in

1963. In 1961, IBM was responsible for 81% of all computers supplied to the government. By 1966, IBM's net shipments to government agencies were negative as a result of returned rental equipment which was replaced by new equipment purchased from IBM's competitors [21].

Table 4
Space-defense computer stock value as percentage of total computer stock value^{a)} (1960–1971, selected years)

Firm	1960	1961	1963	1964	1965	1968	1970	1971
IBM	10	10	8	8	8	7	7	5
Burroughs	18	19	8	5	14	10	23	18
Control Data	5	6	10	13	20	14	16	14
General Electric	0	0	5	4	8	3	5	0
Honeywell	0	2	10	7	9	8	5	4
RCA	15	14	13	15	17	12	15	15
Univac	6	10	11	10	13	15	24	21
Scientific Data	—	—	11	8	4	10	21	28
EMR	—	—	18	9	14	5	24	27
Electronic Assoc.	—	—	—	—	0	12	14	15
Autonetics	13	8	13	22	20	26	15	15
Digital equip.	—	—	9	9	3	2	2	3

Source: A. Gropelli, *The Growth Process in the Computer Industry*, Ph. D. Thesis (New York University, 1970); U. S. Congress, *Inventory of Automatic Data Processing Equipment in Federal Government*, *op. cit.*

a) Value of computer stock computed by using annual inventory data on number of units installed for each computer model and multiplying this figure by estimated monthly rental value for that model.

Market share data demonstrate the importance of space-defense sales for IBM's competitors. Approximately one-fourth of all installed UNIVAC computers were located in space or military agencies by 1970. The comparable percentages for other manufacturers were 23% for Burroughs, 16% for Control Data, 15% for RCA, and 7% for IBM. Newer companies, such as Scientific Data Systems (subsequently, Xerox Data Systems) gained substantially from space-defense demand; in 1971, 28% of all SDS computers were located in space-defense agencies. Other smaller firms such as EMR, and Electronic Associates all had market shares exceeding 15% (table 4) [23].

TECHNOLOGICAL IMPACTS

Although economic impacts are an important part of government influence on high-technology industry, they do not tell the entire story. Essentially, they provide information regarding only the flow of research funds and the

distribution of sales revenue. Yet government programs can also provide important technological support. In the case of the semiconductor and computer industries, the technological benefits for individual firms were significant.

It is important to note that for semiconductor manufacturers, learning by doing on the part of production workers is a relatively unimportant source of learning economies. Most semiconductor learning economies are not an automatic by-product of production, but rather are produced by the deliberate efforts of firms to upgrade production technology. Such economies are generated by improvements in the production process arising from experimentation with the length and temperature of the crystal preparation process, the measures taken to control crystal contamination, the equipment used in assembly and testing, the scheduling and organization of operations, and so on.

The defense and space programs demanded high-quality components from the standpoint of both reliability and performance. The challenging specifications set by the military agencies and NASA accelerated many of the learning economies related to improved production processes. Military and space programs established clear specifications for the types of components they required. This prompted many companies, including newer ones, to develop semiconductors which were at least as good, if not better, than those already under government contract [23].

Whereas military demand accelerated the advance of transistorized component technology, the US space program was dominant in accelerating the development of integrated circuit technology. The development of the Apollo Guidance Computer, a vehicular-borne computer, required extensive miniaturization and extremely high reliability. The MIT Instrumentation Laboratories served as NASA's prime contractor for this project. Twenty-two different component manufacturers participated in the procurement process for the Apollo Guidance Computer. Several of these firms worked with MIT's Instrumentation Laboratories for seven or eight years although their components were never purchased by NASA. The firms remained potential suppliers for such a long period because of the learning experience generated by NASA's demanding technical specifications. NASA's approach to component reliability did not alter commercial production processes in the short run; over the longer run firms did eventually redesign production processes and install new equipment in order to transfer space—defense reliability methods to commercial production.

NASA's efforts to improve the reliability of MOS devices illustrates the technological influence of space program demand on the semiconductor industry. The NASA Microelectronics Reliability Program was instituted to set criteria

for the acceptance of semiconductor devices and required NASA inspection of the production facilities of semiconductor suppliers. All firms interested in supplying MOS devices to NASA had to comply with the NASA Micro-electronic Reliability Program specifications. If a company was unable to meet the acceptance criteria, NASA aided the company in making the necessary changes to meet reliability standards. Major integrated circuit suppliers, such as Motorola, Fairchild, and Harris participated in the program and utilized NASA's technical assistance to improve their processing capabilities [24].

Space-defense demand also influenced technological developments in the computer industry. Work on military and NASA projects accelerated developments in two important areas of computer technology — (1) real-time operating systems and (2) computer system reliability.

The first, and still most famous, real-time computer system was the Air Force's SAGE. The system was set up to protect the United States against a surprise air attack. SAGE consisted of a network of radar-fed computers that continuously analyzed every cubic foot of air space around the US, instantly tracked all approaching aircraft, and decided on an appropriate response. After seven years of development work by MIT's Lincoln Laboratories, with the assistance of IBM and Burroughs, the first of sixteen SAGE centers was completed in 1958. The original computer programs for SAGE consumed 1800 man-years of effort; the total system development cost has been placed at close to \$2 billion.

There is virtual unanimity throughout the data processing community that all subsequent real-time computer systems owe a great deal to SAGE. American Airlines implemented SABRE, its \$30 million real-time seat reservation system in 1964. It is unlikely that SABRE could have been developed at that early date without the experience provided by SAGE.

In the mid-1960's space applications enabled computer manufacturers to further expand their real-time capabilities. Most of the early computer operating systems, which are the set of programs provided to the user by the computer manufacturer for purposes of operating the system, were in a "batch" mode as opposed to a "real-time" mode. For both IBM and Control Data it was NASA's demanding specifications that forced them to advance their capabilities in providing "real time" operating systems.

In fact, IBM's first experience in developing a real-time operating system was with NASA, and the technology developed for that system soon became the industry standard. Control Data's first real-time operating system was also for NASA, at the agency's Langley Research Center. The Langley computer complex is Control Data's largest single installation, and several features of NASA's Langley system have become an integral part of Control Data's standardized real-time operating system [25].

Improved reliability of computer operations is a second area where technological gains were accelerated by space—defense demand. At IBM, Control Data and Univac, NASA's technical specifications prompted significant improvements in the reliability of large computer systems. IBM's major involvement in the space program has been with the computer system at NASA's Manned Spacecraft Center in Houston. This system collects, processes, and sends to Mission Control the information needed to direct manned space missions, such as Apollo and Skylab. The initial reliability specification for this computer system was 41.7 hours of operation without a major breakdown. Originally the IBM system was unable to meet this specification. Special teams of IBM production and test personnel were assigned to work with NASA reliability specialists to correct the problems. By paying extraordinary attention to every aspect of reliability these teams raised the reliability of the real-time computer complex to 100 hours of operation without a catastrophic breakdown, a level which was two and one-half times initial specifications! These new reliability features, which were prompted and financed by NASA, were quickly incorporated into IBM's commercial product line.

MANPOWER IMPACTS

Although economic and technological impacts have been the dominant concern in most efforts to assess the influence of government programs on industry, they do not fully cover the range of possible impacts. Public programs also influence the supply of manpower in industrial firms. These manpower impacts occur in one or more of three ways:

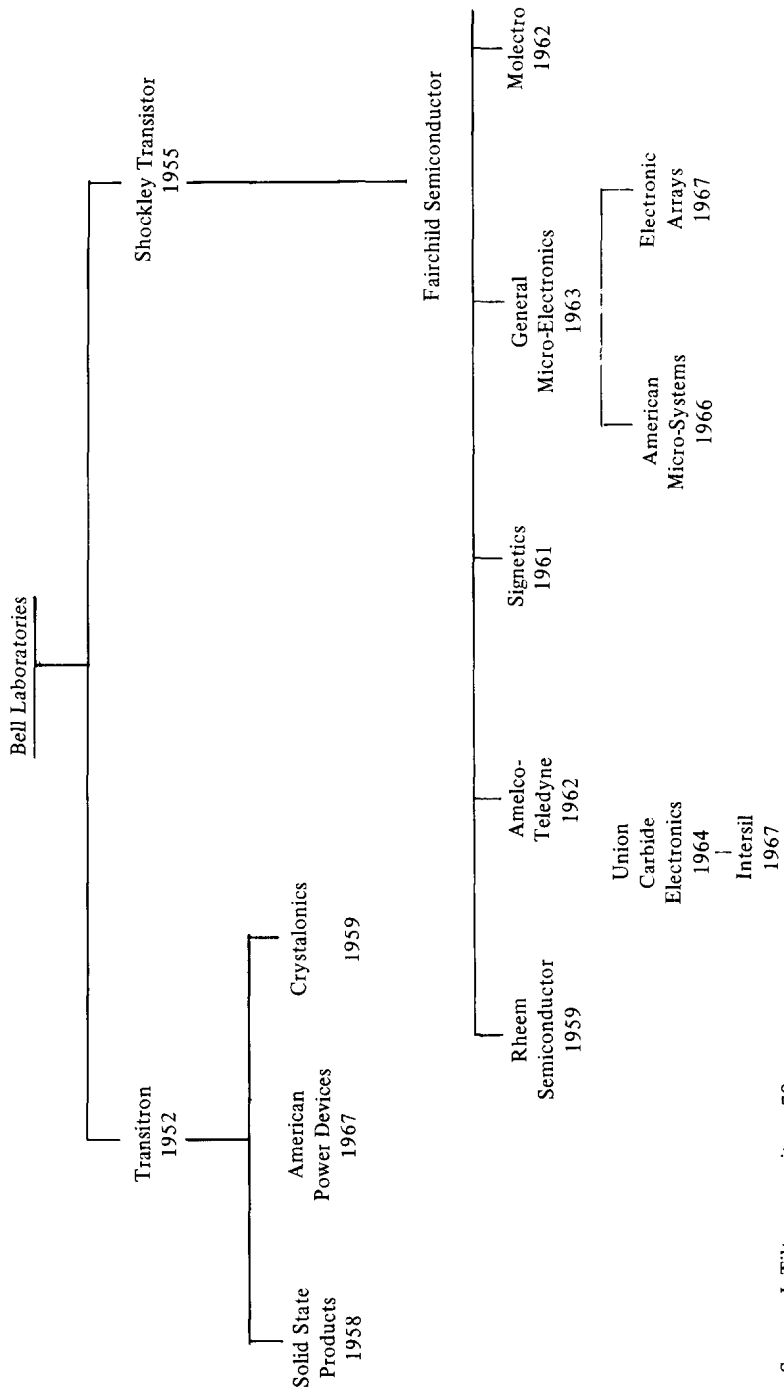
(1) *Direct supply* — A government program can be a direct source of experienced manpower for companies in a particular industry. Personnel may receive training and operating experience in a government program and eventually leave to join a private firm.

(2) *Interfirm mobility* — If government demand enables new firms to enter an industry, a high rate of interfirm mobility may result as personnel leave established older firms to join newer firms.

(3) *Intrafirm mobility* — A firm's work on a government program may provide advanced training for its employees. If the firm seeks to apply the expertise of these employees to other government or commercial projects, a high rate of intrafirm mobility may result.

In the electronics industry the manpower impacts of space defense demand have taken two of these forms. In semiconductors, the willingness of space—defense procurement officials to purchase semiconductor devices from small

Figure 1
Semiconductor Firms Descending from Bell Laboratories, 1952-67



Source: J. Tilton, *op. cit.*, p. 79.

new firms has facilitated high *interfirm mobility*. In contrast, the influence of space—defense programs on computer industry manpower has been to stimulate *intrafirm mobility*. Large space and defense computer systems have provided an important training ground for project managers and engineers, who later applied their knowledge and experience to other large government and commercial computer development projects which were awarded to their firm.

Interfirm mobility in the semiconductor industry includes the movement of personnel among existing firms as well as the departure of engineers and managers from established firms to set up their own companies. Bell Laboratories, Hughes, Philco, RCA, Sylvania, Texas Instruments, and Motorola have all experienced the defection of key employees who set up new firms. Fifteen separate new firms, which were formed between 1952 and 1967, can be traced back to Bell Laboratories. The first new semiconductor firms formed by former Bell Labs personnel were Transitron in 1952 and Shockley Laboratories in 1955. Fairchild Semiconductor, which spun off from Shockley Laboratories in 1955, had five direct spin-offs and four indirect spin-offs over the next 12 years (fig. 1).

Mobility among existing semiconductor firms dates back to the movement of Bell Laboratories employees to Texas Instruments and Sylvania during the early days of the industry's development. During the 1960's, Fairchild was the focal point of much industry mobility. A number of Fairchild executives departed in 1967 to revitalize the National Semiconductor Corporation, which had fallen on hard times. A year later Fairchild recruited the general manager of Motorola's semiconductor division and a number of other Motorola executives [26].

Bell Laboratories has played an important role in the semiconductor industry's high rate of interfirm mobility. As the research arm of AT & T, Bell Laboratories contributed a disproportionately large share of the major semiconductor product and process innovations. Yet AT & T did not use its strong patent position to prevent the entry of new firms to the market; instead, the company adopted a liberal licensing policy and expedited the transfer of technology in a number of ways. AT & T also established the precedent for the industry's tolerant attitude regarding the departure of scientists and engineers to either found new firms or accept positions in competing firms [27].

In the computer industry there is substantial intrafirm mobility in the form of project managers and engineers who move from space—defense divisions to the commercial side of the business. Alumni of IBM's NASA real-time computer complex have gone on to direct the development of major computer systems for the Federal Aviation Administration, the Bank of England and

the City of New York. At Univac, approximately 200 people per year are employed full-time on NASA work. Each year a number of these employees move to another area of the company; consequently Univac estimates that several thousand of its current employees have previously worked on some aspect of NASA systems. One specific example is the project manager of a large computer network developed by Univac for Bell Telephone; he previously served as the firm's Project Manager for NASA's Goddard Space Flight Center. At Control Data, several engineers who acquired real-time simulation experience at NASA's Langley Research Center went on to develop similar applications at Volkswagon and Grumman Aircraft [28].

CONCLUSIONS

The analysis presented in this paper makes it clear that large government programs can play a major role in the growth of high-technology industries. Space and defense programs did, in fact, enable American computer and semiconductor firms to initially assume the leads in their respective fields, positions they have maintained since. It should be noted, however, that despite the many positive impacts of these programs, governmental involvement in these areas has not been free from criticism.

Critics have complained that the US space and defense programs created imbalances in the nation's supply of scientific and technological manpower. In some areas of engineering and science, these federal programs helped to produce a supply of engineers and scientists which now exceeds current demand. Because NASA and defense provided substantial funding and devoted specific efforts to attract and educate more scientists and engineers, these agencies assumed difficult responsibilities. The instability of government funding and sudden changes in program plans have had serious consequences for the affected scientists and engineers.

There are those who contend that the government's reduction in funding for science and engineering constitutes a serious breach of trust. The message for public program managers is clear: Large public programs cannot become deeply involved in influencing technological manpower flows for the purposes of accomplishing specific agency objectives and avoid responsibility for subsequent negative impacts which may result. Indeed, the experience of the US space and defense programs suggests that government managers must be more sensitive to the longer term manpower consequences of their programs [29]. What is the more general significance of this case study for the role of large public programs in advancing science and technology? The technological

impacts of space and defense illustrate that the managers of a large government program face a major dilemma in making scientific and technical decisions. If they are to ensure the accomplishment of specified program objectives, they must plan to use only currently available scientific knowledge and technology. They must be technically conservative and focus on exploiting the existing state-of-the-art rather than carrying out revolutionary scientific advances within their program. Consequently, it is unlikely that public programs will originate major innovations, although they can play an important role in technological advance. Given the constraints imposed by specific program objectives, it is unrealistic to expect evaluations to show that a government program deserves sole or direct credit for generating important scientific developments.

While public programs usually concentrate on exploiting the existing technology, they do not necessarily receive a "free ride" from other institutions, such as industrial firms or universities. Programs such as space and defense have increased the speed at which the state-of-the-art advances. Although the space program was not a "pioneer" in the computer and semiconductor industries, it did accelerate technological trends toward extensive miniaturization and high reliability. Commercial products such as pocket calculators and digital watches, promoted as space-age innovations (complete with "Apollo" trademark), did not result from the space program, but they probably have become available to consumers much earlier because of NASA [30].

Public programs can also be influential in overcoming the resistance to technological change which inevitably limits early acceptance of new products and managerial methods. In the computer industry the "demonstration effect" provided by the space program's computers greatly increased the acceptance of the computer by the business world. The widely publicized use of computers for their industrial applications made business executives more willing to use computers for their industrial applications.

Finally, we should not expect the technological impacts of public programs to produce economic returns in the short run. The process of converting a significant technological advance into new products or processes is complex and time consuming. Studies of major twentieth innovations indicate that the time interval between an invention and its commercial use averages between ten and twenty years. Although such studies indicate that the average time interval between invention and innovation for government-supported innovations is shorter than the average for privately supported innovations, the lag is still substantial [31].

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